Road Network Navigation in Any Pose

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1. Course Content

Basic

- Start the road network navigation feature, and master using the roadnet_nav visualization panel to send a goal pose so that, combined with the road network planning feature, the robot reaches the goal in any pose.

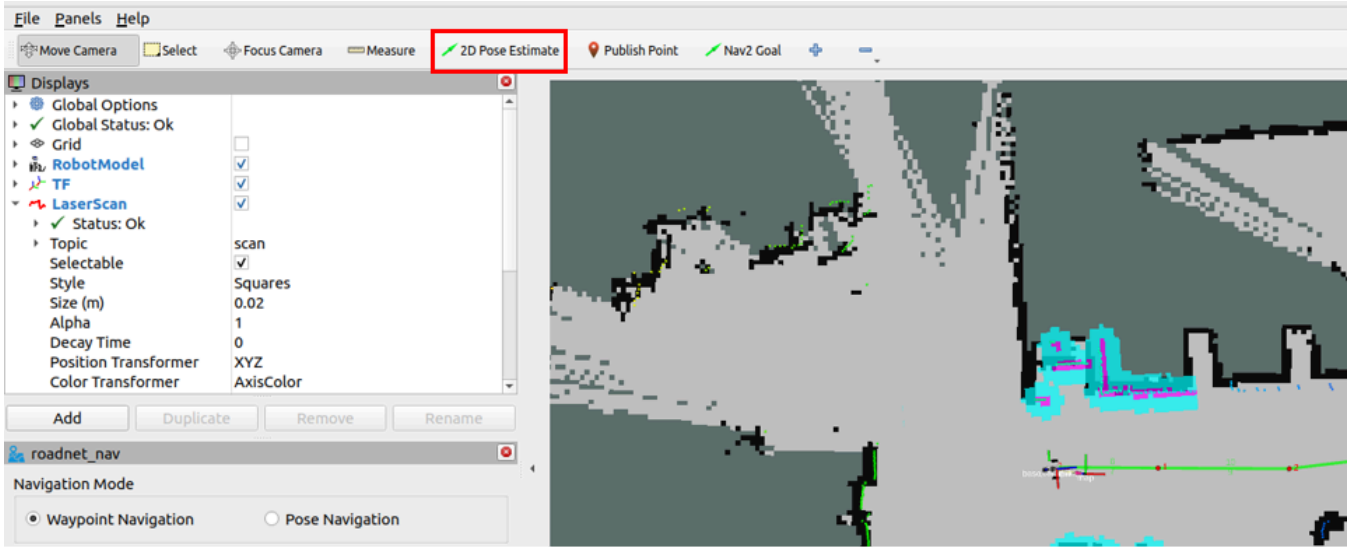
Advanced

- Master the method of requesting navigation through the action service interface
- Understand the implementation logic of the program source code

2. Preparation

```
ros2 launch yahboomcar_bringup roadnet_nav.launch.py
```

- Click **2D Pose Estimate** to initialize the robot's pose on the map



3. Road Network Planning and Navigation in Any Pose

- Here we use road network navigation by publishing to the `/road_net_nav` topic in the terminal. This

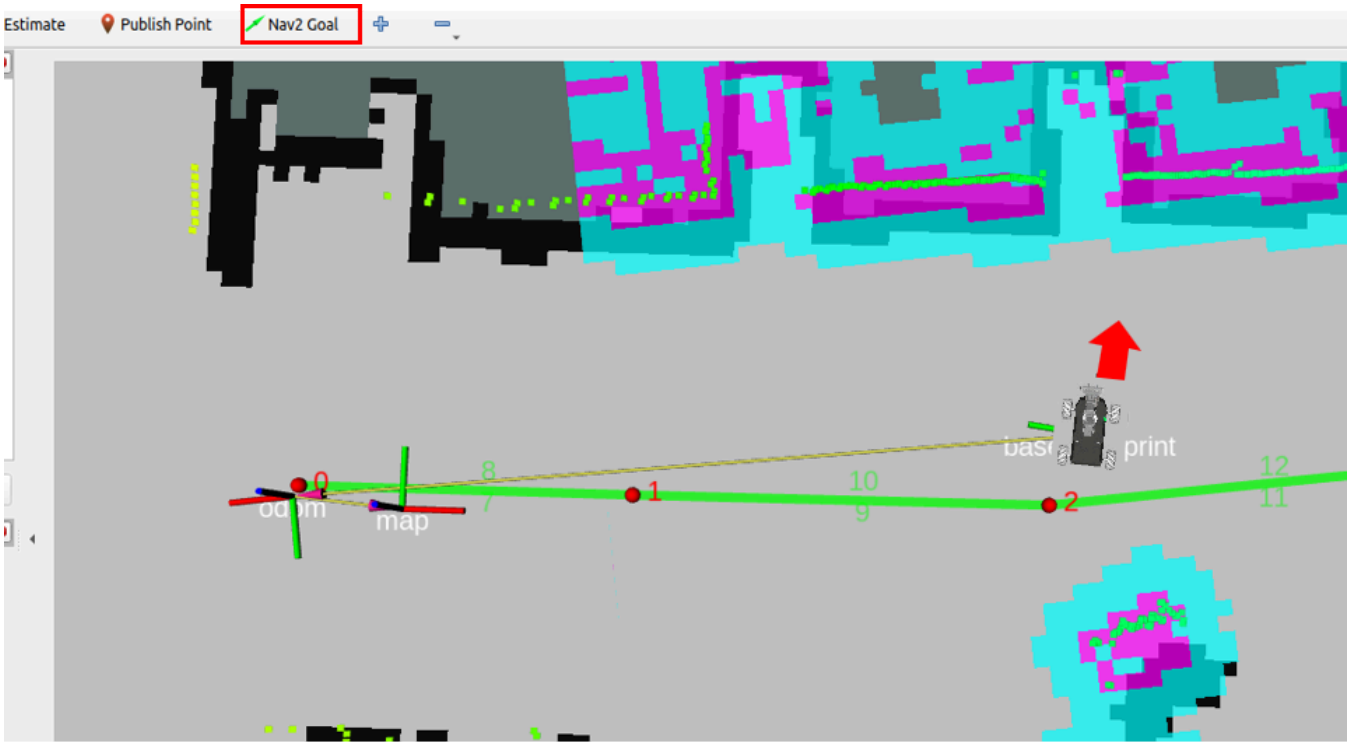
topic can accept a `PoseStamped` goal pose, travel along the road network to the vicinity of the goal pose, and then precisely adjust to the goal pose. The topic publishing format is as follows:

```
ros2 topic pub -1 /road_net_nav geometry_msgs/PoseStamped "{
    pose: {
      position: {x: xx, y: xx, z: 0.0},
      orientation: {x: 0.0, y: 0.0, z: xx, w: xx}
  }
}"
```

> **Note**
>
> A pose consists of position and orientation. Here, for a ground mobile robot, the position has only planar x, y coordinates, and the orientation is only rotation about its own z axis. In pose:
>
> - **position: {x: xx, y: xx, z: 0.0}** is the position information
> - **orientation: {x: 0.0, y: 0.0, z: xx, w: xx}** is the body rotation information described by a quaternion

3.1 Case Demonstration

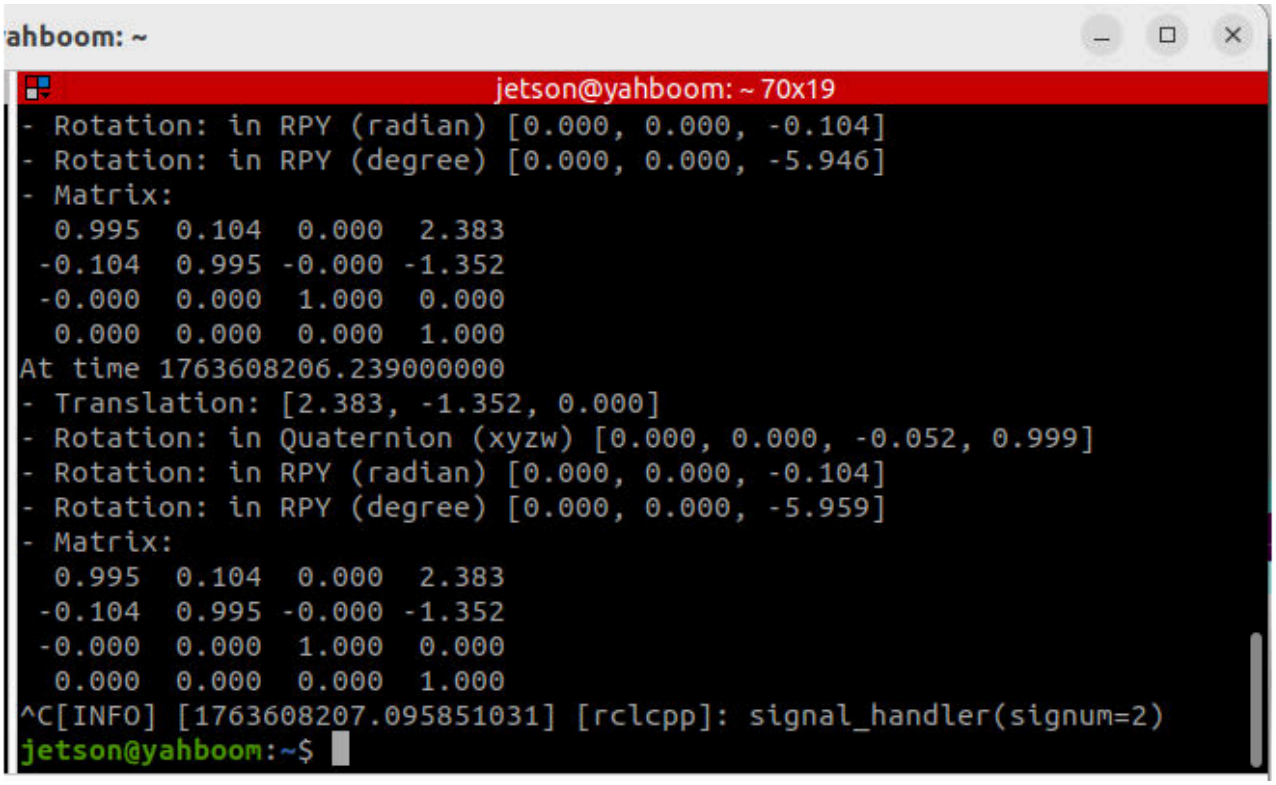
- Use the **Nav2 Goal** tool to first navigate to a pose point



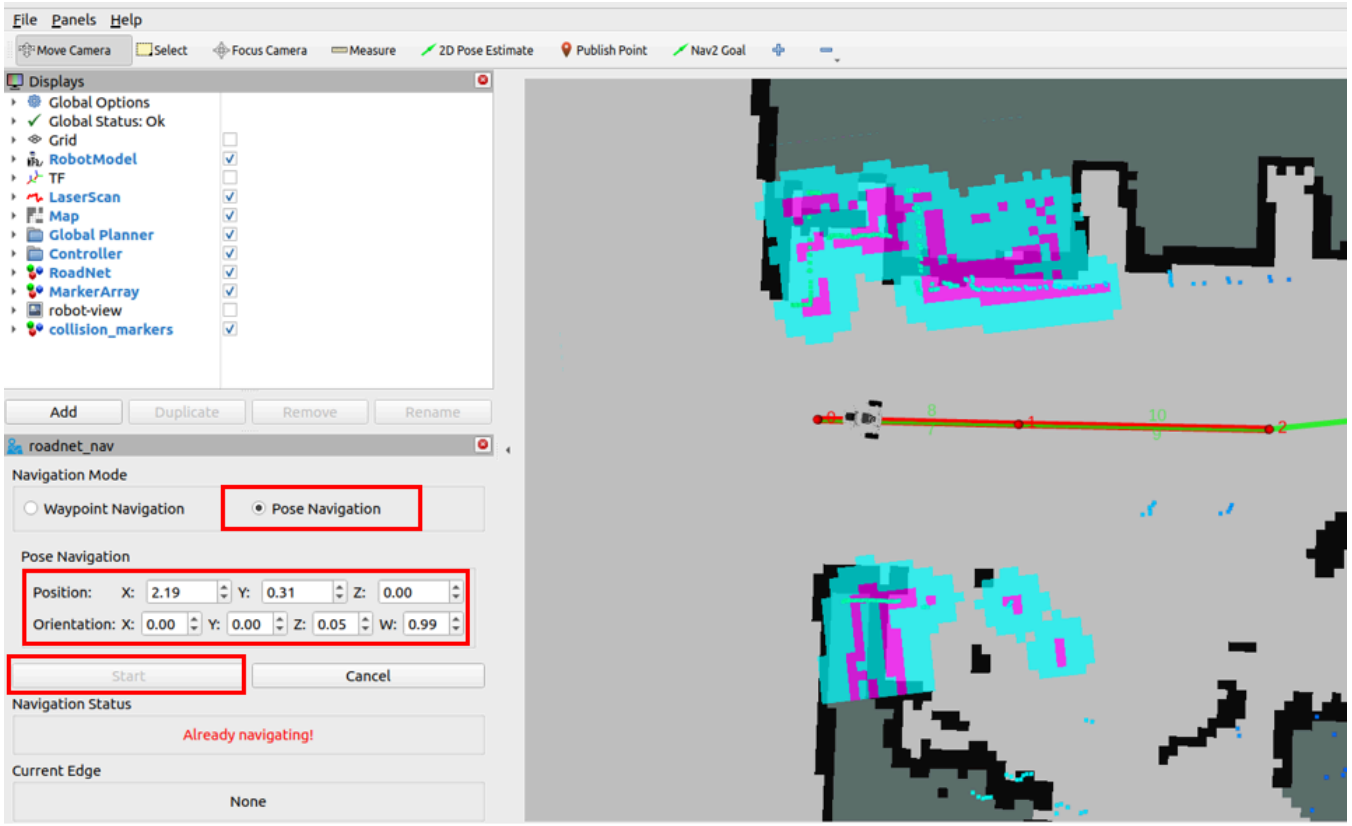
- Use the following command to obtain the robot's current coordinates in the map

```
ros2 run tf2_ros tf2_echo map base_footprint
```

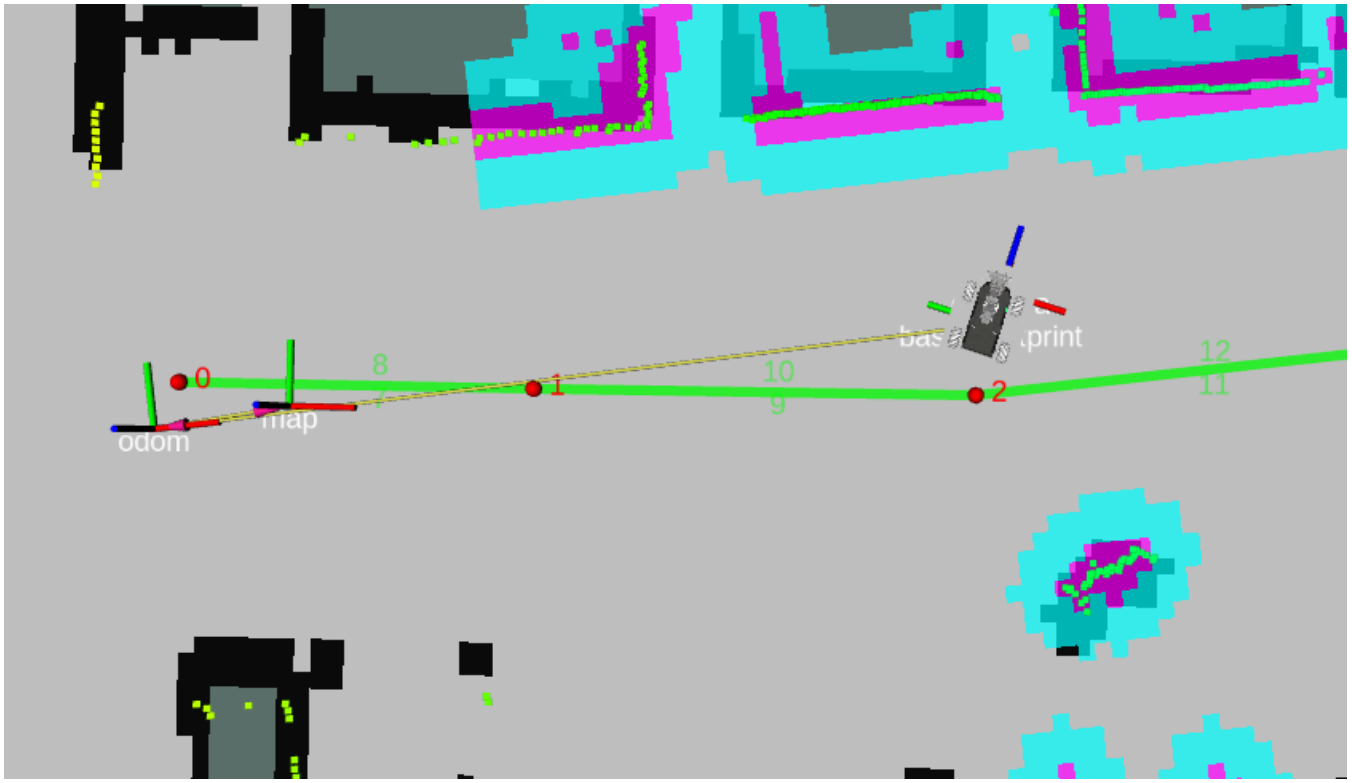
Here `Translation` and `Rotation: in Quaternion (xyzw)` are the robot's pose in the global map.



- Then we let the robot return to its initial position, and select **Pose Navigation** in the **roadnet_nav** panel, fill in the pose just queried, and click **Start** to begin navigation



- After that, the robot will first travel along the road network to a waypoint near the goal point, then navigate along a road network edge, and finally adjust to the goal pose.



3.2 Requesting Navigation Through the Action Service Interface

- Replace the pose parameter with the pose queried earlier; its function is the same as that in the **roadnet_nav** panel

```
ros2 action send_goal --feedback /compute_route_and_follow roadnet/action/RoadNetToGoal \
"{
  use_poses: true,
  goal: {
    header: {frame_id: 'map'},
    pose: {position: {x: 2.19, y: 0.31, z: 0.0}, orientation: {z: 0.05, w: 0.99}}
  }
}"
```

4. Source Code Analysis

- Refer to the source code analysis in the 【Waypoint-to-Waypoint Planning and Navigation】 chapter lesson; the `computeRouteAndFollow` function contains the implementation of the pose navigation part

5. Parameter Debugging

- Refer to the parameter debugging in the 【Waypoint-to-Waypoint Planning and Navigation】 chapter lesson; the parameter file is the same